These are notes I'm taking for Christian Gromoll's Probability Theory class. The primary text is *Probability with Martingales* by Williams.

Dictionary between Measure Theory and Probability

We will assume a lot of measure theory background, so we will instead discuss the dictionary between terms from probability and terms from measure theory. It's important to note that the measure spaces in probability theory are probability measure spaces (i.e., they have measure 1).

Measure Theory	Probability
Set X	Sample space Ω
σ-Algebra Σ	σ-Field $\mathcal{F}$
Measurable subsets $A \in \Sigma$	Events $F \in \mathcal{F}$
Measure $\mu(A)$	Probability $P(F)$
Almost Everywhere (a.e.)	Almost Surely (a.s.)
$\limsup_{n\to\infty} E_n$	$\{E_n$ infinitely often$\}$, $\{E_n$ i.o.$\}$
$\liminf_{n\to\infty} E_n$	$\{E_n$ eventually$\}$, $\{E_n$ ev$\}$
Borel Measurable Function $f: X \to \mathbb{R}$	Random Variable $X: \Omega \to \mathbb{R}$
Characteristic function χ_A	Indicator function 1_A
Integral $\int f \, d\mu$	Expectation $E[X]$
Pushforward measure $f_*\mu$	Law Λ_X
$\mu(S)$	$P(\omega \in S)$
$f_*\mu(S')$	$P(X \in S')$

Remark: In probability, rather than the non-calligraphic L^p spaces (whose elements are equivalence classes of almost-everywhere equal functions), one uses the calligraphic $\mathcal{L}^p$, whose elements are the functions themselves. This does not fundamentally change anything in particular (as probabilists are only concerned with almost-everywhere equivalence regardless), but it is useful to note.

Important Theorems

σ-Algebras and Measures

Definition: Let S be a set, and let $\mathcal{D}$ be a collection of subsets. We call $\mathcal{D}$ a π-system on S if it is closed under finite intersection.

We call $\mathcal{D}$ a λ-system if

- $S \in \mathcal{D}$;
- for any $A, B \in \mathcal{D}$ with $A \subseteq B$, one has $B \setminus A \in \mathcal{D}$;
- if $(A_n)_n \subseteq \mathcal{D}$ is such that $\bigcup_n A_n = A$, then $A \in \mathcal{D}$.

For an arbitrary collection of subsets $\mathcal{I}$, we write $\pi(\mathcal{I})$ for the π-system generated by $\mathcal{I}$ and $\lambda(\mathcal{I})$ for the λ-system generated by $\mathcal{I}$.

Proposition: A collection Σ of subsets of S is a σ-algebra if and only if Σ is both a π-system and a λ-system.

Proof. Let Σ be a π-system and a λ-system. If $E, F \in \Sigma, (E_n)_n \subseteq \Sigma$, then

$$E^c = S \setminus E$$

$$E \cup F = S \setminus (E^c \cap F^c).$$

Therefore, $G_n = E_1 \cup \cdots \cup E_n \in \Sigma$, with $\bigcup_n G_n = \bigcup_k E_k$, so that $\bigcup_k E_k \in \Sigma$. Finally, we have $\bigcap_k E_k = \left(\bigcup_k E_k^c\right)^c$, meaning that Σ is a σ-algebra. □

Theorem (Dynkin π-λ Theorem)**:** If $\mathcal{I}$ is a π-system, then $\lambda(\mathcal{I}) = \sigma(\mathcal{I})$.

Proof. To prove the theorem, we will use a classic technique by defining a "truth set" on which a desired statement holds, then showing that truth set is the desired σ-algebra.

Define

$$\mathcal{D}_1 = \{B \in \lambda(\mathcal{I}) : B \cap C \in \lambda(\mathcal{I}) \text{ for all } C \in \mathcal{I}\}.$$

We observe that $\mathcal{I}$ is a π system, so that $\mathcal{I} \subseteq \mathcal{D}_1$. Furthermore,

- $S \in \mathcal{D}_1$;
- for any $B_1, B_2 \in \mathcal{D}_1$ with $B_1 \subseteq B_2$, we have that for any given $C \in \mathcal{I}$, $(B_2 \setminus B_1) \cap C = (B_2 \cap C) \setminus (B_1 \cap C)$, so since $\lambda(\mathcal{I})$ is a λ-system, it follows that $B_2 \setminus B_1 \in \mathcal{D}_1$;
- if $(B_n)_n \subseteq \mathcal{D}_1$ increases to B, then for each $C \in \mathcal{I}$, one has $(B_n \cap C)_n \nearrow B \cap C$, so that $B \cap C \in \lambda(\mathcal{I})$, meaning $B \in \mathcal{D}_1$.

Thus, $\mathcal{D}_1$ is a λ-system containing I, so since $\mathcal{D}_1 \subseteq \lambda(\mathcal{I})$ by definition, one has $\mathcal{D}_1 = \lambda(\mathcal{I})$.

Next, define

$$\mathcal{D}_2 = \{A \in \lambda(\mathcal{I}) : B \cap A \in \lambda(\mathcal{I}) \text{ for all } B \in \lambda(\mathcal{I})\}.$$

From above, we have that $\mathcal{D}_2$ contains I. Similarly to above, we can prove that $\mathcal{D}_2$ inherits the λ-system structure from $\lambda(\mathcal{I})$, so that $\mathcal{D}_2 = \lambda(\mathcal{I})$. Yet, this means that $\mathcal{D}_2 = \lambda(\mathcal{I})$, so $\lambda(\mathcal{I})$ is a π-system. □

Corollary: If two probability measures μ_1 and μ_2 agree on a π-system $\mathcal{I}$, then they agree on the σ-algebra $\sigma(\mathcal{I})$.

Theorem (Carathéodory Extension Theorem)**:** Let S be a set, Σ_0 an algebra of subsets on S, and $\Sigma = \sigma(\Sigma_0)$. If $\mu_0 \colon \Sigma_0 \to [0, \infty]$ is a countably additive map, then there exists a measure μ on (S, Σ) such that $\mu = \mu_0$ on Σ_0.

Furthermore, if $\mu_0(S) < \infty$, then the extension is unique.

Convergence Theorems

We recall some of the useful convergence theorems from measure theory, specializing to the case with measure 1 and rewriting in the language of probability.

Theorem (Monotone Convergence Theorem)**:** If $(X_n)_n$ is a sequence of real-valued random variables increasing pointwise almost surely to a real-valued random variable X, then

$$E[X] = \lim_{n \to \infty} E[X_n].$$

Theorem (Fatou's Lemma)**:** If $(X_n)_n$ is a sequence of positive-valued random variables, then

$$E\left[\liminf_{n \to \infty} X_n\right] \leq \liminf_{n \to \infty} E[X_n]$$
$$\limsup_{n \to \infty} E[X_n] \leq E\left[\limsup_{n \to \infty} E[X_n]\right].$$

Theorem (Dominated Convergence Theorem)**:** Let $(X_n)_n$ be a sequence of complex-valued random variables converging pointwise almost surely to a complex-valued random variable X. If there exists some positive-

valued integrable random variable Y such that $|X_n| \leq Y$ for all n, then $(X_n)_n \xrightarrow{\mathcal{L}^1} X$. That is,

$$E[|X_n - X|] \to 0,$$

so that $E[X_n] \to E[X]$.

Theorem (Scheffé Convergence Theorem)**:** If $(X_n)_n$ is a sequence of (real valued) integrable random variables converging to an integrable random variable X almost surely, and $E(|X_n|) \to E(|X|)$, then $(X_n)_n \xrightarrow{\mathcal{L}^1} X$.

Remark: The Scheffé convergence theorem is proven by splitting into positive and negative components.

Independence

Definition: We say sub-σ-algebras $\mathcal{G}_1, \mathcal{G}_2, \ldots$ of $\mathcal{F}$ are *independent* if for any $G_i \in \mathcal{G}_i$ with distinct i, one has

$$P\Big(G_{i_1} \cap \cdots \cap G_{i_n}\Big) = \prod_{k=1}^{n} P\Big(G_{i_k}\Big).$$

We say random variables $X_1, X_2, \ldots$ are independent if their produced σ-algebras $\sigma(X_1), \sigma(X_2), \ldots$ are independent.

Events $E_1, E_2, \ldots$ are independent if their σ-algebras $\mathcal{E}_n$, where

$$\mathcal{E}_n = \{\emptyset, E_n, E_n^c, \Omega\},$$

are independent.

Theorem: If X and Y are independent random variables, then

$$E[XY] = E[X]E[Y].$$

Proof. Bootstrap from simple functions. □

Asymptotic Results

Theorem (First Borel–Cantelli Lemma)**:** Let $(E_n)_n$ be a family of events such that $\sum_n P(E_n) < \infty$. Then,

$$\begin{aligned} P\left(\limsup_{n\to\infty} E_n\right) &= P(E_n \text{ i.o.}) \\ &= 0. \end{aligned}$$

Proof. Let

$$G_m = \bigcup_{n=m}^{\infty} E_n.$$

Then, $G_m \searrow \limsup E_n$, and one has

$$\begin{aligned} P(\limsup E_n) &\leq P(G_m) \\ &\leq \sum_{n=m}^{\infty} P(E_n). \end{aligned}$$

Taking $m \to \infty$ yields our desired result. □

Theorem (Second Borel–Cantelli Lemma)**:** Suppose $(E_n)_n$ is a sequence of independent events. Then, if $\sum_n P(E_n) = \infty$, one has

$$P(\limsup E_n) = 1.$$

Proof. Recall that $(\limsup E_n)^c = \liminf E_n^c$. If $p_n = P(E_n)$, then one has

$$P\left(\bigcap_{n\geq m} E_n^c\right) = \prod_{n\geq m}(1-p_n).$$

Notice that if $x \geq 0$, then $1 - x \leq e^{-x}$. Therefore, we have

$$\begin{aligned}\prod_{n\geq m}(1-p_n) &\leq \exp\left(-\sum_{n\geq m} p_n\right)\\ &= 0,\end{aligned}$$

where the latter equality emerges from the fact that $\sum_n p_n = \infty$.

Therefore $P(\liminf E_n^c) = 0$, meaning $P(\limsup E_n) = 1$. □

Definition: Let $X_1, X_2, \ldots$ be random variables, and let

$$\begin{aligned}\mathcal{T}_n &= \sigma(X_{n+1}, X_{n+1}, \ldots)\\ \mathcal{T} &= \bigcap_n \mathcal{T}_n.\end{aligned}$$

We call $\mathcal{T}$ the *tail* σ-algebra of the sequence $(X_n)_n$.

We notice that $\mathcal{T}$ contains many important such events:

(i) $F_1 = \{\lim_k X_k \text{ exists}\}$;

(ii) $F_2 = \{\sum_k X_k \text{ exists}\}$;

(iii) $F_3 = \left\{\lim_k \frac{\sum_{i=1}^k X_i}{k} \text{ exists}\right\}$.

The most useful fact about the tail σ-algebra for a sequence of independent random variables is the *Kolmogorov Zero-One Law*, which says that tail events occur with either probability 0 or probability 1.

Theorem: Let $(X_n)_n$ be a sequence of independent random variables, and let $\mathcal{T}$ be the tail σ-algebra. Then,

(i) $F \in \mathcal{T}$ implies $P(F) = 0$ or $P(F) = 1$;

(ii) if ξ is a $\mathcal{T}$-measurable random variables, then there is some constant c in $[-\infty, \infty]$ such that $P(\xi = c) = 1$.

Proof.

(i) Let

$$\begin{aligned}\mathcal{X}_n &= \sigma(X_1, \ldots, X_n)\\ \mathcal{T}_n &= \sigma(X_{n+1}, X_{n+1}, \ldots).\end{aligned}$$

First, we observe that $\mathcal{X}_n$ and $\mathcal{T}_n$ are independent. This follows from the fact that the class $\mathcal{K}$ defined by

$$\mathcal{K} = \{\{\omega : X_i(\omega) \leq x_i\} : x_i \in \mathbb{R} \cup \{\infty\}, 1 \leq i \leq n\}$$

is a π-system that generates $\mathcal{X}_n$. The class $\mathcal{J}$ defined by

$$\mathcal{J} = \left\{\left\{\omega : X_j(\omega) \leq x_j\right\} : x_j \in \mathbb{R} \cup \{\infty\}, r \in \mathbb{N}, n+1 \leq j \leq n+r\right\}$$

is a π-system that generates $\mathcal{T}_n$. By the assumption that the sequence $(X_n)_n$ is independent, we have that $\mathcal{K}$ and $\mathcal{J}$ are independent. Therefore, the corresponding σ-algebras are independent as well.

Since $\mathcal{T} \subseteq \mathcal{T}_n$, it follows that $\mathcal{X}_n$ and $\mathcal{T}$ are independent. Next, if we let $\mathcal{X}_\infty = \sigma(X_1, X_2, \dots)$, then we have that $\mathcal{K}_\infty = \bigcup_n \mathcal{X}_n$ is a π-system that generates $\mathcal{X}_\infty$. Furthermore, since $\mathcal{K}_\infty$ and $\mathcal{T}$ are independent, it follows that $\mathcal{X}_\infty$ and $\mathcal{T}$ are independent.

Yet, we have that $\mathcal{T} \subseteq \mathcal{X}_\infty$, meaning that $\mathcal{T}$ is independent of $\mathcal{T}$. Therefore, if $F \in \mathcal{T}$, we have that

$$\begin{aligned} P(F) &= P(F \cap F) \\ &= P(F)P(F), \end{aligned}$$

whence $P(F) = 0$ or $P(F) = 1$.

(ii) For every $x \in \mathbb{R}$, we have that $P(\xi \leq x) = 0$ or $P(\xi \leq x) = 1$.

Let

$$c = \sup\{x : P(\xi \leq x) = 0\}.$$

Then, if $c = -\infty$, we have that $P(\xi = -\infty) = 1$, while if $c = \infty$, then $P(\xi = \infty) = 1$. Else, if c is finite, then $P(\xi \leq c - 1/n) = 0$ for all n, whence

$$\begin{aligned} P\left(\bigcup_n \{\xi \leq c - 1/n\}\right) &= 0 \\ &= P(\xi < c), \end{aligned}$$

while $P(\xi \leq c + 1/n) = 1$ for all n, so that

$$\begin{aligned} P\left(\bigcap_n \{\xi \leq c + 1/n\}\right) &= 1 \\ &= P(\xi \leq c). \end{aligned}$$

Thus, $P(\xi = c) = 1$.

□

Inequalities

Theorem (Chebyshev's Inequality)**:** Let Z be a measurable function on $(\Omega, \mathcal{F}, P)$, and let $g\colon \mathbb{R} \to [0, \infty]$ be a Borel measurable increasing function. Then,

$$P(Z \geq c) \leq \frac{1}{c} E[Z]$$

for all $c \in \mathbb{R}$.

Proof. We observe that

$$\begin{aligned} E[Z] &= \int Z \, dP \\ &\geq \int_{\{Z \geq c\}} Z \, dP \\ &\geq \int_{\{Z \geq c\}} dP \\ &= P(Z \geq c). \end{aligned}$$

□

Theorem (Jensen's Inequality)**:** Let $c\colon G \to \mathbb{R}$ be a convex function on an open interval, and let X be a random variable such that

- $E[|X|] < \infty$;
- $P(X \in G) = 1$;
- $E(|c(X)|) < \infty$.

Then, $E[c(X)] \geq c(E[X])$.

Proof. Since c is convex, we observe that whenever $u < v < w$, we have $\frac{c(v)-c(u)}{v-u} \leq \frac{c(w)-c(v)}{w-v}$. In particular, since for any $x < y$ in G there are $z, w \in G$ with $z > w > y$, it follows that c is continuous as we have $\left|c(y) - c(x)\right| \leq \frac{c(z)-c(w)}{z-w}\left|y - x\right|$.

Therefore, the monotone derivatives

$$(D_- c)(v) = \lim_{u \nearrow v} \frac{c(v) - c(u)}{v - u}$$
$$(D_+ c)(v) = \lim_{w \searrow v} \frac{c(w) - c(v)}{w - v}$$

exist and have $(D_- c)(v) \leq (D_+ c)(v)$.

These functions are increasing, and for every $v \in G$ and any $m \in [(D_- c)(v), (D_+ c)(v)]$, we have $c(x) \geq m(x - v) + c(v)$ for all $x \in G$.

Therefore, if $\mu = E[X]$, we have $c(X) \geq m(X - E[X]) + c(E[X])$ for any $m \in [(D_- c)(E[X]), (D_+ c)(E[X])]$.

Taking expectations gives the desired result. □

Theorem (Hölder's Inequality)**:** If p and q are conjugate exponents, then for any $f \in \mathcal{L}^p$ and $g \in \mathcal{L}^q$, we have $fg \in \mathcal{L}^1$ with $\|fg\|_{\mathcal{L}^1} \leq \|f\|_{\mathcal{L}^p} \|g\|_{\mathcal{L}^q}$.

Theorem (Minkowski's Inequality)**:** For any $f, g \in \mathcal{L}^p$, we have $\|f + g\|_{\mathcal{L}^p} \leq \|f\|_{\mathcal{L}^p} + \|g\|_{\mathcal{L}^p}$.

Theorem (Inequality of $\mathcal{L}^p$ Norms)**:** Suppose $1 \leq p \leq r < \infty$, and $Y \in \mathcal{L}^r$. Then, $Y \in \mathcal{L}^p$, and we have $\|Y\|_{\mathcal{L}^p} \leq \|Y\|_{\mathcal{L}^r}$.

Theorem (Cauchy–Schwarz Inequality)**:** If X and Y are in $\mathcal{L}^2$, then $XY \in \mathcal{L}^1$ and satisfy $E[|XY|] \leq \|X\|_{\mathcal{L}^2} \|Y\|_{\mathcal{L}^2}$.

Martingales

The idea behind martingales is intimately connected to the notion of conditional expectation.

Theorem: Let $(\Omega, \mathcal{F}, P)$ be a probability measure, and let X be a random variable with $E[|X|] < \infty$. Let $\mathcal{G}$ be some sub-σ-algebra of $\mathcal{F}$. Then, there exists a random variable Y such that

(i) Y is $\mathcal{G}$-measurable;

(ii) $E[|Y|] < \infty$;

(iii) for every $G \in \mathcal{G}$, we have

$$\int_G Y \, dP = \int_G X \, dP.$$

The random variable Y is unique up to almost sure equality, and we call it a version of the *conditional expectation* $E[X|\mathcal{G}]$.

Proof. Define a measure μ on $\mathcal{G}$ by taking, for any $G \in \mathcal{G}$,

$$\mu(G) = \int_G X \, dP.$$

Since X is integrable, $\mu(G)$ is finite, and we have $\mu \ll P$ on $\mathcal{G}$. Therefore, the Radon–Nikodym derivative

$\frac{d\mu}{dP}$ exists, is integrable, $\mathcal{G}$-measurable, and definitionally satisfies what we desire from a conditional expectation. □

There is a method by using the Hilbert Projection Theorem to construct the conditional expectation by first defining it for $\mathcal{L}^2$-random variables, then approximating by truncation. The conditional expectation satisfies the following properties commensurate with how we defined the measure μ, though parts (j) and (k) are proven via bootstrapping.

Theorem: The conditional expectation $E[X|\mathcal{G}]$ satisfies the following properties:

(a) $E[E[X|\mathcal{G}]] = E[X]$;

(b) if X is $\mathcal{G}$-measurable, then $E[X|\mathcal{G}] = E[X]$;

(c) $E[a_1X_1 + a_2X_2|\mathcal{G}] = a_1E[X_1|\mathcal{G}] + a_2E[X_2|\mathcal{G}]$;

(d) if $X \geq 0$, then $E[X|\mathcal{G}] \geq 0$;

(e) if $0 \leq X_n \nearrow X$, then $E[X_n|\mathcal{G}] \nearrow E[X|\mathcal{G}]$;

(f) if $X_n \geq 0$, then $E[\liminf X_n|\mathcal{G}] \leq \liminf E[X_n|\mathcal{G}]$ and $E[\limsup X_n|\mathcal{G}] \geq \limsup E[X_n|\mathcal{G}]$;

(g) if $|X_n(\omega)| \leq V(\omega)$ for all n, $E[V] < \infty$, and $X_n \to X$ almost surely, then $E[X_n|\mathcal{G}] \to E[X|\mathcal{G}]$;

(h) if $c\colon \mathbb{R} \to \mathbb{R}$ is convex with $E[|c(X)|] < \infty$, then $E[c(X)|\mathcal{G}] \geq c(E[X|\mathcal{G}])$;

(i) if $\mathcal{H}$ is a sub-σ-algebra of $\mathcal{G}$, then $E[E[X|\mathcal{G}]|\mathcal{H}] = E[X|\mathcal{H}]$;

(j) if Z is $\mathcal{G}$-measurable and bounded, then $E[ZX|\mathcal{G}] = ZE[X|\mathcal{G}]$;

(k) if $\mathcal{H}$ is independent of $\sigma(\sigma(X), \mathcal{G})$, then $E[X|\sigma(\mathcal{G}, \mathcal{H})] = E[X|\mathcal{G}]$.

Definition and Examples of a Martingale

Definition: Let $(\Omega, \mathcal{F}, P)$ be a probability space. A (discrete) filtration $\{\mathcal{F}_n : n \geq 0\}$ is a sequence of sub-σ-algebras $\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \cdots \subseteq \mathcal{F}$. We will define

$$\mathcal{F}_\infty = \sigma\left(\bigcup_{n=0}^{\infty} \mathcal{F}_n\right).$$

The tuple $(\Omega, \mathcal{F}, \{\mathcal{F}_n\}_n, P)$ is called a *filtered probability space*.

Definition: A stochastic process $W = (W_n)_{n\geq 0}$ is a sequence of $\mathcal{F}$-measurable random variables. A stochastic process is said to be *adapted* to a filtration $\{\mathcal{F}_n\}_n$ if each W_n is $\mathcal{F}_n$-measurable.

Example: If W is a stochastic process, and

$$\mathcal{F}_n = \sigma(W_0, \ldots, W_n)$$

is the corresponding produced σ-algebra, then W is adapted to the filtration $\{\mathcal{F}_n\}_n$. Such a filtration is dubbed the natural filtration on a given stochastic process.

Definition: A stochastic process X is called a *martingale* relative to some filtration $\{\mathcal{F}_n\}$ if

(i) X is adapted to the filtration;

(ii) each X_n in X satisfies $E[|X_n|] < \infty$;

(iii) $E[X_n|\mathcal{F}_{n-1}] = X_{n-1}$ for all $n \geq 1$.

If $X_{n-1} \geq E[X_n|\mathcal{F}_{n-1}]$, then we call the process a *supermartingale*, and if $X_{n-1} \leq E[X_n|\mathcal{F}_n]$, then we call the process a *submartingale*. Notice that submartingales are increasing in expectation while supermartingales are decreasing in expectation.

Equivalently, we say that $(X_n)_n$ is a submartingale if

$$E[X_n - X_{n-1}|\mathcal{F}_{n-1}] \geq 0$$

and $(X_n)_n$ is a supermartingale if

$$E[X_n - X_{n-1}|\mathcal{F}_{n-1}] \leq 0.$$

Example (Sums of Independent Random Variables)**:** Suppose $(X_n)_n$ is a sequence of independent, integrable, centered random variables. Let $S_0 = 0$, $S_n = X_1 + \cdots + X_n$, and $\mathcal{F}_n = \sigma(X_1, \ldots, X_n)$, where $\mathcal{F}_0 = \{\emptyset, \Omega\}$. Then, for any $n \geq 1$, we have that

$$\begin{aligned} E[S_n|\mathcal{F}_{n-1}] &= E[S_{n-1}|\mathcal{F}_{n-1}] + E[X_n|\mathcal{F}_{n-1}] \\ &= S_{n-1} + E[X_n] \\ &= S_{n-1}. \end{aligned}$$

The first equality follows from linearity, the second from the fact that S_{n-1} is $\mathcal{S}_{n-1}$-measurable and that X_n is independent of $\mathcal{F}_{n-1}$, and the third from the fact that the X_n are centered.

Example (Products of Independent Random Variables)**:** Suppose $(X_n)_n$ is a sequence of independent integrable random variables with mean 1. Then, defining

$$\begin{aligned} M_0 &= 1 \\ M_n &= X_1 \cdots X_n \\ \mathcal{F}_0 &= \{\emptyset, \Omega\} \\ \mathcal{F}_n &= \sigma(X_1, \ldots, X_n), \end{aligned}$$

we have by a similar procedure that

$$\begin{aligned} E[M_n|\mathcal{F}_{n-1}] &= E[M_{n-1}X_n|\mathcal{F}_{n-1}] \\ &= M_{n-1}E[X_n|\mathcal{F}_{n-1}] \\ &= M_{n-1}E[X_n] \\ &= M_{n-1}. \end{aligned}$$

Example (Accumulating Information)**:** Let $\{\mathcal{F}_n : n \in \mathbb{N}\}$ be a filtration, and let $\xi \in \mathcal{L}^1(\Omega)$. Let $M_n = E[\xi|\mathcal{F}_n]$. By the tower property, we have

$$\begin{aligned} E[M_n|\mathcal{F}_{n-1}] &= E[\xi|\mathcal{F}_n|\mathcal{F}_{n-1}] \\ &= E[\xi|\mathcal{F}_{n-1}] \\ &= M_{n-1}. \end{aligned}$$

Martingale Transforms and Stopping Times

Definition: We call a process $C = (C_n)_{n\geq 1}$ *predictable* (or *previsible*) if C_n is $\mathcal{F}_{n-1}$-measurable.

We may consider C_n as a stake on game n in some repeated game. The winnings at time n are given by $C_n(X_n - X_{n-1})$. The total winnings up to time n are then given by

$$\begin{aligned} Y_n &= \sum_{k=1}^{n} C_k(X_k - X_{k-1}) \\ &=: (C \cdot X)_n. \end{aligned}$$

This is known as the *martingale transform* of X by C. It is the discrete analogue of a stochastic integral $\int C\, dX$. We define $C_0 = 0$.

Theorem (No Beating a Fair Game)**:**

(i) Let C be a bounded positive predictable process. That is, there is some $K \geq 0$ such that $|C_n(\omega)| \leq K$ for all n and ω. Let X be a supermartingale. Then, $C \cdot X$ is a supermartingale with $(C \cdot X)_0 = 0$.

(ii) If C is a bounded predictable process, and X is a martingale, then $C \cdot X$ is a martingale with $(C \cdot X)_0 = 0$.

(iii) In (i) and (ii), we may replace boundedness with the condition that $C_n \in \mathcal{L}^2$ assuming that $X \in \mathcal{L}^2$.

Proof.

(i) Let $Y = C \cdot X$. Since C_n is bounded, positive, and $\mathcal{F}_{n-1}$-measurable, we have

$$\begin{aligned} E[Y_n - Y_{n-1}|\mathcal{F}_{n-1}] &= C_n E[X_n - X_{n-1}|\mathcal{F}_n] \\ &\leq 0. \end{aligned}$$

(ii) Replace ≤ 0 with $= 0$ in the above expression.

(iii) This result follows from applying the definition of the conditional expectation emerging from the formulation via the Hilbert projection theorem.

□

Definition: We say a map $T : \Omega \to \{0, 1, 2, \ldots, \infty\}$ is a *stopping time* if $\{T \leq n\} := \{\omega : T(\omega) \leq n\}$ is contained in $\mathcal{F}_n$ for all $n \leq \infty$. Equivalently, $\{T = n\} \in \mathcal{F}_n$ for all $n \leq \infty$.

By convention, we have $\inf(\emptyset) = \infty$ and $\sup(\emptyset) = 0$.

Example: Suppose $(A_n)_n$ is an adapted process, and let $B \in \mathcal{F}$. Let

$$T = \inf\{n \geq 0 : A_n \in B\}.$$

Then, T is a stopping time, seeing as

$$\begin{aligned} \{T \leq n\} &= \bigcup_{k \leq n} \{A_k \in B\} \\ &\in \mathcal{F}_n. \end{aligned}$$

Consider a supermartingale X, and let T be a stopping time. The betting process is given by betting 1 unit at each time and quitting at time T. The stake process is then given by $C_n^{(T)} = 1_{\{n \leq T\}}$, where 1_E is the indicator function. Then,

$$C_n^{(T)} = \begin{cases} 1 & n \leq T(\omega) \\ 0 & \text{else.} \end{cases}$$

The martingale transform is then given by $\left(C^{(T)} \cdot X\right)_n = X_{T \wedge n} - X_0$. The *stopped process* is then given by

$$X_n^T(\omega) = X_{T(\omega) \wedge n}(\omega),$$

so that $C^{(T)} \cdot X = X^T - X_0$. Notice that $C^{(T)}$ is bounded and positive, and is predictable since $C_n^{(T)}$ can only be 0 or 1, and $\left\{C_n^{(T)} = 0\right\} = \{T \leq n-1\} \in \mathcal{F}_{n-1}$ for any $n \geq 1$. Thus, since $C^{(T)}$ is a bounded positive predictable process, we have the following.

Theorem: If X is a supermartingale, and T is a stopping time, then the stopped process X^T is a supermartingale. In particular, $E[X_{T \wedge n}] \leq E[X_0]$.

Similarly, if X is a martingale, and T is a stopping time, then X^T is a martingale. In particular, $E[X_{T \wedge n}] = E[X_0]$.

Consider a simple symmetric random walk on $\mathbb{Z}$ with stopping time

$$T = \inf\{n : X_n = 1\}.$$

It can be shown that $P(T < \infty) = 1$, as we will show later. However, even though $E[X_{T\wedge n}] = E[X_0] = 0$ for every n, we have $E[X_T] = 1$. Our goal is to understand when $E[X_T] = E[X_0]$.

Theorem (Doob's Optional Stopping Theorem)**:**

(a) Let T be a stopping time, and X a supermartingale. Then, X_T is integrable with $E[X_T] \leq E[X_0]$ if any of the following conditions hold:

(i) T is bounded — that is, for some $N \in \mathbb{N}$, $T(\omega) \leq N$ for all ω;

(ii) X is bounded (that is, uniformly bounded) and T is almost surely finite;

(iii) $E[T] < \infty$ and X has bounded increments, in that there exists $K \geq 0$ such that

$$|X_n(\omega) - X_{n-1}(\omega)| \leq K$$

for all $n \in \mathbb{N}$ and $\omega \in \Omega$.

(b) If any of (i)–(iii) hold, and X is a martingale, then $E[X_T] = E[X_0]$.

Proof.

(a) We know that $X_{T\wedge n}$ is integrable with $E[X_{T\wedge n} - X_0] \leq 0$. To obtain (i), we may take $n = N$, so that $T \wedge N = T$. To obtain (ii), we may let $n \to \infty$ and use the dominated convergence theorem. Finally, to obtain (iii), we observe that

$$\begin{aligned} |X_{T\wedge n} - X_0| &= \left| \sum_{k=1}^{T\wedge n} (X_k - X_{k-1}) \right| \\ &\leq KT. \end{aligned}$$

Since $E[KT] < \infty$, we may use dominated convergence yet again by taking $n \to \infty$.

(b) Apply the result in (a) to X and $-X$.

□

Corollary:

(a) Suppose M is a martingale with bounded increments, C is a bounded predictable process, and T is a stopping time with $E[T] < \infty$. Then, $E\big[(C \cdot M)_T\big] = 0$.

(b) If X is a positive supermartingale and T is an almost surely finite stopping time, then $E[X_T] \leq E[X_0]$.

Knowing when $E[T] < \infty$ is a valuable criterion to understand when X_T is a martingale.

Lemma: Suppose T is a stopping time such that for some $N \in \mathbb{N}$ and some $\varepsilon_0 > 0$, we have for every $n \in \mathbb{N}$ that

$$P(T \leq n + N | \mathcal{F}_n) \geq \varepsilon$$

almost surely. Then, $E[T] < \infty$.

Proof. Our goal is to show that $P(T = \infty) = 0$. For this, consider the tail of all such kN. We will show that $P(T > kN) \leq (1-\varepsilon)^k$. Automatically, we have

$$\begin{aligned} P(T > N) &= 1 - P(T \leq N) \\ &= 1 - P(T \leq N | \mathcal{F}_0) \\ &= 1 - \varepsilon. \end{aligned}$$

Now, if we assume that the relation $P(T > kN) \leq (1-\varepsilon)^k$ holds, then we find that

$$\begin{aligned} P(T > (k+1)N) &= P(T > (k+1)N \cap T > kN) \\ &= E\Big[1_{\{T>(k+1)N\}} 1_{\{T>kN\}}\Big] \end{aligned}$$

$$\begin{aligned}
&= E\Big[E\Big[1_{\{T>kN\}}1_{\{T>(k+1)N\}}|\mathcal{F}_{kN}\Big]\Big] \\
&= E\Big[1_{\{T>kN\}}E\Big[1_{\{T>(k+1)N\}}|\mathcal{F}_{kN}\Big]\Big] \\
&= E\Big[(1-\varepsilon)1_{\{T>kN\}}\Big] \\
&= (1-\varepsilon)P(T>kN) \\
&= (1-\varepsilon)^{k+1}.
\end{aligned}$$

Notice that

$$\begin{aligned}
P(T=\infty) &= \lim_{k\to\infty} P(T>kN) \\
&= \lim_{k\to\infty} (1-\varepsilon)^k \\
&= 0.
\end{aligned}$$

Therefore, $T<\infty$ almost surely, so that $E[T]<\infty$. □

Our next consideration is understanding hitting times for a simple random walk. Suppose $(X_n)_n$ is a sequence of independent and identically distributed random variables with $P(X=1)=P(X=0)=1/2$. If $S_0=0$ and $S_n=X_1+\cdots+X_n$, then we set

$$T=\inf\{n : S_n=1\},$$

with the natural filtration $\mathcal{F}_n=\sigma(S_0,\ldots,S_n)$. Notice that if X is the random variable with identical distribution to the X_n, then$E\Big[e^{\theta X}\Big]=\cosh(\theta)$, so that $E\Big[(\operatorname{sech}(\theta))e^{\theta X_n}\Big]=1$. In particular, M_n^θ, given by

$$M_n^\theta=(\operatorname{sech}(\theta))^n e^{\theta S_n}$$

is a martingale as it is a product of mean-1 i.i.d. random variables.

Martingale Convergence Theorem

Suppose $(X_n)_n$ is some stochastic process, and let $Y=C\cdot X$ be the corresponding transform by some predictable process C. We will consider a strategy as follows: after choosing two numbers $a<b$

- wait until $X<a$;
- bet one unit until X is above b
- repeat this betting strategy forever.

We inductively define

$$\begin{aligned}
C_1 &= 1_{\{X_0<a\}} \\
C_n &= 1_{\{C_{n-1}=1\}}1_{\{X_{n-1}<b\}}+1_{\{C_{n-1}=0\}}1_{\{X_{n-1}<a\}}
\end{aligned}$$

to formally define this betting strategy.

Definition: The *upcrossing number* $U_N[a,b](\omega)$ counts the largest k such that

$$0\le s_1<t_1<s_2<\cdots<s_k<t_k\le N$$

such that $X_{s_i}(\omega)<a$ and $X_{t_i}(\omega)>b$ for each $1\le i\le k$.

In this context, the upcrossing number essentially defines our winnings. Notice that for each N, we have

$$Y_N(\omega)\ge(b-a)U_N[a,b](\omega)+(X_N(\omega)-a)_-,$$

where

$$f_-=-\min(f,0)$$

is the negative part of the function f.

Lemma (Doob Upcrossing Lemma)**:** Let X be a supermartingale, and let $U_N[a,b]$ be the number of upcrossings of $[a,b]$ by time N. Then,

$$(b-a)E[U_N[a,b]] \leq E[(X_N - a)_-].$$

Proof. Since C is predictable, bounded, and positive with $Y = C \cdot X$, it follows that Y is a supermartingale, so that $E[Y_N] \leq 0$. □

Corollary: Let X be a uniformly bounded supermartingale in $\mathcal{L}^1$. That is,

$$\sup_n E[|X_n|] < \infty.$$

Let $a, b \in \mathbb{R}$ with $a < b$. Then, $U_\infty[a,b] = \lim_{N\to\infty} U_N[a,b]$ satisfies

$$(b-a)E[U_\infty[a,b]] \leq |a| + \sup_n E[|X_n|],$$

whence $P(U_\infty[a,b] = \infty) = 0$.

Proof. As a general fact, we have that $(X_N - a)_- \leq |a| + |X_N|$. Therefore, we have

$$\begin{aligned}(b-a)E[U_N[a,b]] &\leq |a| + E[|X_N|] \\ &\leq |a| + \sup_n E[|X_n|].\end{aligned}$$

Using the monotone convergence theorem then gives our desired result. □

Theorem ('Forward' Convergence Theorem)**:** Let X be a uniformly bounded supermartingale in $\mathcal{L}^1$. Then, $X_\infty := \lim_n X_n$ exists and is finite a.s.

Proof. For definiteness, we will write $X_\infty(\omega) = \limsup_n X_n(\omega)$. Let

$$\begin{aligned}\Lambda &:= \{\omega : X_n(\omega) \text{ does not converge to a limit in } [-\infty,\infty]\} \\ &= \left\{\omega : \liminf_n X_n(\omega) < \limsup_{n\to\infty} X_n(\omega)\right\} \\ &= \bigcup_{\substack{a,b\in\mathbb{Q}\\ a<b}} \left\{\omega : \liminf_n X_n(\omega) < a < b < \limsup_n X_n(\omega)\right\} \\ &=: \bigcup_{\substack{a,b\in\mathbb{Q}\\ a<b}} \Lambda_{a,b}.\end{aligned}$$

Yet, we have that $\Lambda_{a,b} \subseteq \{\omega : U_\infty[a,b](\omega) = \infty\}$, so that $P(\Lambda_{a,b}) = 0$. Thus, $P(\Lambda) = 0$, whence $X_\infty := \lim_n X_n$ exists almost surely in $[-\infty,\infty]$.

By Fatou's Lemma, we have

$$\begin{aligned}E[|X_\infty|] = E\left[\liminf_n |X_n|\right] &\leq \liminf_n E[|X_n|] \\ &\leq \sup_n E[|X_n|] \\ &< \infty.\end{aligned}$$

Thus $P(X_\infty = \infty) = 0$. □

Square-Integrable Martingales

When we specialize to the case of martingales in $\mathcal{L}^2$, we may use a variety of useful properties of the geometry of $\mathcal{L}^2$ space. This enables, for instance, the proof of the three series theorem and the vaunted strong law of large numbers for i.i.d. random variables.

We work with martingales $(M_n)_n$ that are bounded in $\mathcal{L}^2$, in the sense that

$$\sup_n E\Big[M_n^2\Big] < \infty.$$

Since, for any $s \leq t \leq u \leq v$, we have

$$E[M_v | \mathcal{F}_u] = M_u,$$

it follows that

$$\langle M_t - M_s, M_v - M_u \rangle = 0.$$

Therefore, writing

$$M_n = M_0 + \sum_{k=1}^{n} (M_k - M_{k-1}),$$

we have that M_n is expressed as the sum of pairwise orthogonal terms, so that by the Pythagorean theorem,

$$E\Big[M_n^2\Big] = E\Big[M_0^2\Big] + \sum_{k=1}^{n} E\Big[(M_k - M_{k-1})^2\Big].$$

Immediately we get the following result.

Theorem: Let M be a martingale for which $M_n \in \mathcal{L}^2$ for all n. Then, M is bounded in $\mathcal{L}^2$ if and only if

$$\sum_{k=1}^{\infty} E\Big[(M_k - M_{k-1})^2\Big] < \infty.$$

When this occurs, we have $M_n \to M_\infty$ a.s. and in $\mathcal{L}^2$.

Proof. Our main job will be to show that $M_n \to M_\infty$ a.s. and in $\mathcal{L}^2$. That M is bounded in $\mathcal{L}^2$ emerges immediately from the above decomposition of M_n into orthogonal components. So, M_∞ exists as $\mathcal{L}^2$-boundedness implies $\mathcal{L}^1$-boundedness, meaning that by the Forward Convergence Theorem, $M_\infty := \lim_n M_n$ exists.

By the Pythagorean theorem, we have

$$E\Big[(M_{n+1} - M_n)^2\Big] = \sum_{k=n+1}^{n+r} E\Big[(M_k - M_{k-1})^2\Big].$$

Letting $r \to \infty$ and applying Fatou's lemma, we have

$$E\Big[(M_\infty - M_n)^2\Big] \leq \sum_{k=n+1}^{\infty} E\Big[(M_k - M_{k-1})^2\Big].$$

We may take the right-hand side to be arbitrarily small by assumption, so that

$$\lim_n E\Big[(M_\infty - M_n)^2\Big] = 0.$$

□

Theorem: Suppose $(X_k)_k$ is a sequence of independent random variables such that $E[X_k] = 0$ for each k and $\sigma_k^2 := \text{var}(X_k) < \infty$. Then,

(a) $\sum_k \sigma_k^2 < \infty$ implies $\sum_k X_k$ converges a.s.;

(b) if the variables $(X_k)_k$ are uniformly bounded, then $\sum_k X_k$ converges a.s. implies that $\sum_k \sigma_k^2 < \infty$.

Proof. Define

$$\begin{aligned}
\mathcal{F}_n &= \sigma(X_1, \dots, X_n) \\
M_n &= X_1 + X_2 + \cdots + X_n \\
A_n &= \sum_{k=1}^{n} \sigma_k^2 \\
N_n &= M_n^2 - A_n,
\end{aligned}$$

with $A_0 = 0$ and $N_0 = 0$.

(a) We know that M is a martingale since it is the sum of mean-zero independent random variables. Moreover,

$$\begin{aligned}
E\Big[(M_k - M_{k-1})^2\Big] &= E\Big[X_k^2\Big] \\
&= \sigma_k^2,
\end{aligned}$$

so by the Pythagorean theorem,

$$\begin{aligned}
E\Big[M_n^2\Big] &= \sum_{k=1}^{n} \sigma_k^2 \\
&= A_n.
\end{aligned}$$

If $\sum_k \sigma_k^2 < \infty$, then M is $\mathcal{L}^2$-bounded, so $\lim_n M_n$ exists a.s.

(b) Since X_k is independent of $\mathcal{F}_{k-1}$, we have that

$$\begin{aligned}
E\Big[(M_k - M_{k-1})^2 | \mathcal{F}_{k-1}\Big] &= E\Big[X_k^2 | \mathcal{F}_{k-1}\Big] \\
&= E\Big[X_k^2\Big] \\
&= \sigma_k^2.
\end{aligned}$$

Since M_{k-1} is $\mathcal{F}_{k-1}$-measurable, we have that

$$\begin{aligned}
\sigma_k^2 &= E\Big[M_k^2 | \mathcal{F}_{k-1}\Big] - 2M_{k-1}E[M_{k-1} | \mathcal{F}_{k-1}] + M_{k-1}^2 \\
&= E\Big[M_k^2 | \mathcal{F}_{k-1}\Big] - M_{k-1}^2.
\end{aligned}$$

Yet, this gives that N is a martingale. Let $0 < c < \infty$, and define $T = \inf\{r : |M_r| > c\}$. Then, the stopped martingale N^T is a martingale, so for every n, we have

$$\begin{aligned}
E\Big[N_n^T\Big] &= E\Big[\Big(M_n^T\Big)^2\Big] - E[A_{T \wedge n}] \\
&= 0.
\end{aligned}$$

Since $|M_T - M_{T-1}| = |X_T| \leq K$ if T is finite, it follows that $\Big|M_n^T\Big| \leq K + c$ for all n, whence $E[A_{T \wedge n}] \leq (K + c)^2$.

Since $\sum_n X_n$ converges a.s., the partial sums are bounded, and it must be the case that for some c, $P(T = \infty) > 0$. Thus, $A_\infty := \sum_{k=1}^{\infty} \sigma_k^2 < \infty$.

□

Remark: We note that the convergence of $\sum_k X_k$ is a tail event, so if $(X_n)_n$ is a sequence of independent random variables that is uniformly bounded, then if the probability that the partial sums of $\sum_k X_k$ are bounded is strictly positive, it follows that the sum $\sum_k X_k$ is almost surely.

Example: If $(a_n)_n$ is a sequence of real numbers, and $(\varepsilon_n)_n$ is a sequence of i.i.d. random variables with

$$P(\varepsilon_n = \pm 1) = 1/2,$$

then the results above show that $\sum_n \varepsilon_n a_n$ converges if and only if $\sum_n a_n^2 < \infty$.

We will prove the following result by using a special technique known as symmetrization.

Lemma: Suppose $(X_n)_n$ is a sequence of independent random variables uniformly bounded by a constant $0 \leq K < \infty$ and $\sum_n X_n$ converges almost surely. Then, $\sum_n E[X_n]$ converges and $\sum_n \text{var}(X_n) < \infty$.

Proof. If each X_n has mean zero, then we use the above theorem.

Else, we will construct a family of mean zero independent random variables that are uniformly bounded. We will write $(\Omega', \mathcal{F}', P')$ be an exact copy of $(\Omega, \mathcal{F}, P)$, and similarly let $(X_n')_n$ a copy of $(X_n)_n$.

$$\begin{aligned}(\Omega^*, \mathcal{F}^*, P^*) &= (\Omega, \mathcal{F}, P) \times (\Omega', \mathcal{F}', P') \\ Y_n(\omega, \omega') &= X_n(\omega) \\ Y_n'(\omega, \omega') &= X_n'(\omega') \\ Z_n(\omega, \omega') &= X_n(\omega) - X_n'(\omega').\end{aligned}$$

The family of random variables $\{Y_n, Y_n' : n \in \mathbb{N}\}$ is independent on $(\Omega^*, \mathcal{F}^*, P^*)$, and both Y_n and Y_n' have the same P^*-distribution as the P-distribution of X_n.

In particular, $(Z_n)_n$ is a mean-zero sequence of independent random variables on $(\Omega^*, \mathcal{F}^*, P^*)$ with $|Z_n(\omega\omega')| \leq 2K$ and $\text{var}(Z_n) = 2\sigma_n^2$.

Suppose

$$G = \left\{ \omega \in \Omega : \sum_n X_n(\omega) \text{ converges} \right\},$$

and similarly define G' on Ω'. Then, $P(G) = P(G') = 1$, so the rectangle set has probability $P(G \times G') = P(G)P(G') = 1$. Since $\sum_n Z_n(\omega, \omega')$ converges on $G \times G'$, we have that $P^*(\sum_n Z_n \text{ converges}) = 1$. Since we have a sum of uniformly bounded mean-zero independent random variables that converges a.s., it follows that $\sum_n \sigma_n^2 < \infty$, and $\sum_n (X_n - E[X_n])$ converges almost surely. Therefore, $\sum_n E[X_n]$ converges. □

These above two characterizations of convergence for sums of square-integrable random variables will be used to prove the celebrated Kolmogorov Three-Series Theorem.

Theorem (Kolmogorov Three-Series Theorem)**:** Let $(X_n)_n$ be a sequence of independent random variables. Then, $\sum_n X_n$ converges almost surely if and only if for some K, the following series converge:

(i) $\displaystyle\sum P(|X_n| > K)$;

(ii) $\displaystyle\sum E\Big[X_n 1_{\{|X_n| \leq K\}}\Big]$;

(iii) $\displaystyle\sum \text{var}\Big(X_n 1_{|X_n| \leq K}\Big)$.